Smart Pantry App

MAD 700 Report

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MOBILE APP DEVELOPMENT 700

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

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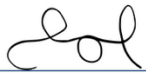
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Introduction

Problem Statement

A lot of food gets wasted because it ends up in hidden corners and gets completely forgotten about. By the time it is found, it has already expired and needs to be thrown away resulting in food waste

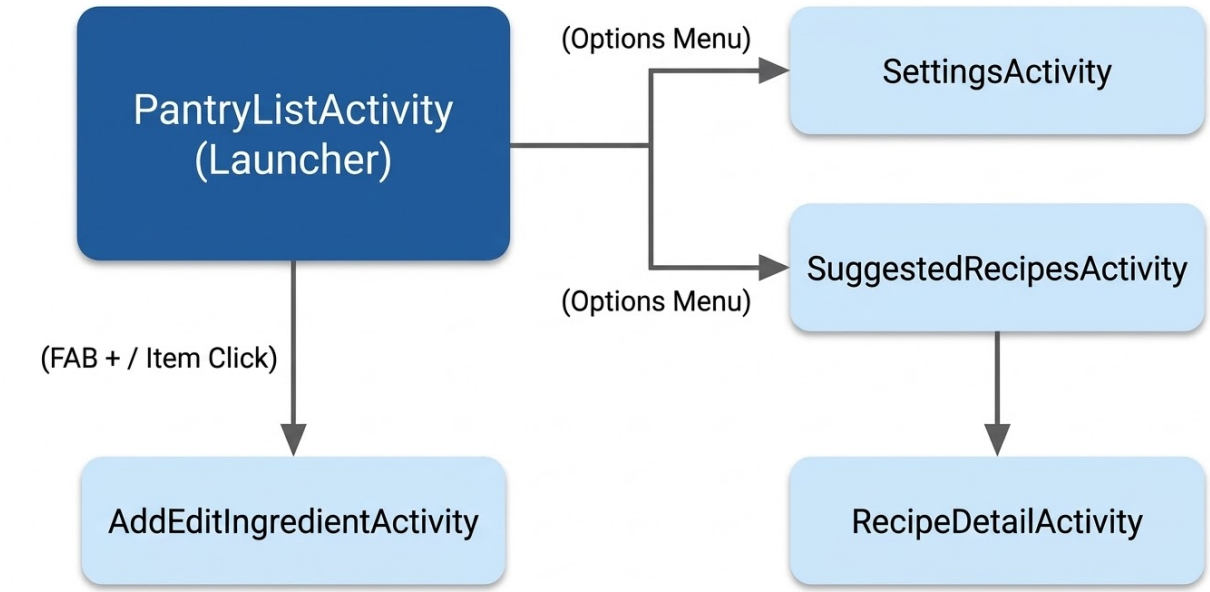
Proposed Solution

I created a smart pantry app that helps people keep track of all the food inside the pantry. The app includes a notification feature that reminds the user when food is about to expire. This gives them a 2nd chance to use up food before it expires. The app provides recipe suggestions based off the current pantry list to help users make use of what's available instead of having to go shopping to make food. It maximizes the use of available ingredients.

Repo link

http://github.com/Muhluri765/smart-pantry-app

System Design



PantryListActivity

The main screen on the app. Displays list of food items in the pantry

SettingsActivity

Activity is accessed through a kebab menu. Users can change settings. Currently you can change the day the expiry notifications begin and change between imperial and metric units

SuggestedRecipesActivity

Accessed through a button at the top of PantryListActivity. Presents a list of recipe suggestions based off available ingredients inside the pantry

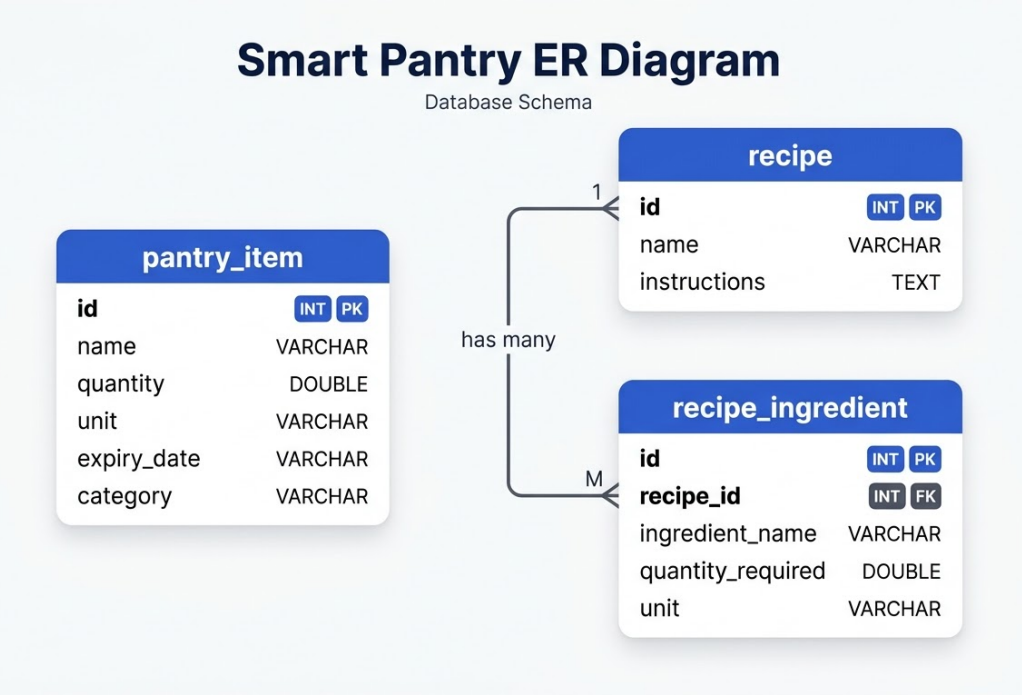
RecipeDetailActivity

Displayed when a user taps on one of the recipes displayed on the SuggestedRecipesActivity. This activity displays the ingredients and instructions to make a specific recipe.

AddEditIngredientActivity

When you tap the “+” button at the bottom of PantryListActivity, it will open this activity. This activity allows you to fill in the details of an ingredient (name, number of units, expiry date & category).

Entity Relationship Diagram



Application Screenshots

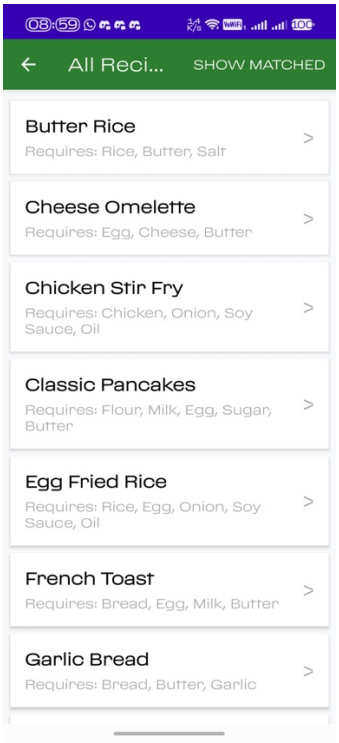


Figure 1: SuggestedRecipes showing list of possible recipes with current ingredients

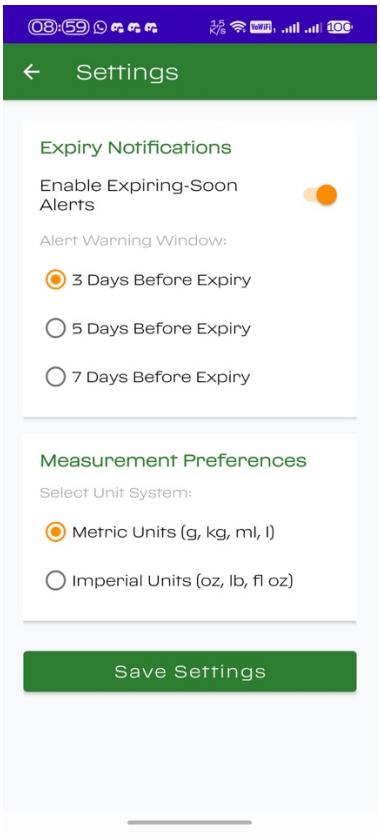


Figure 2: SettingsActivity. Allows user to change notification timing and switch between imperial and metric units

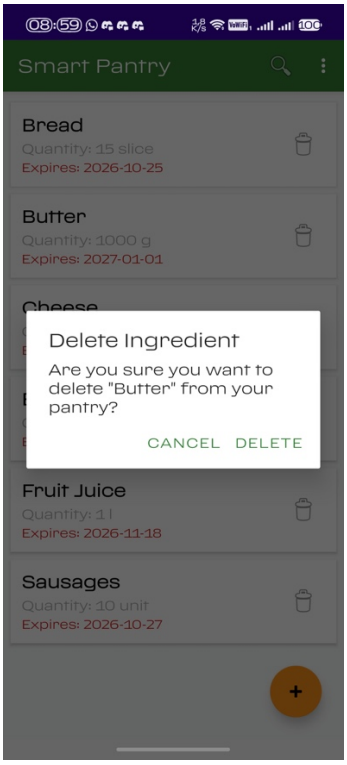


Figure 3: popup that appears to confirm if user wants to delete an ingredient from the list

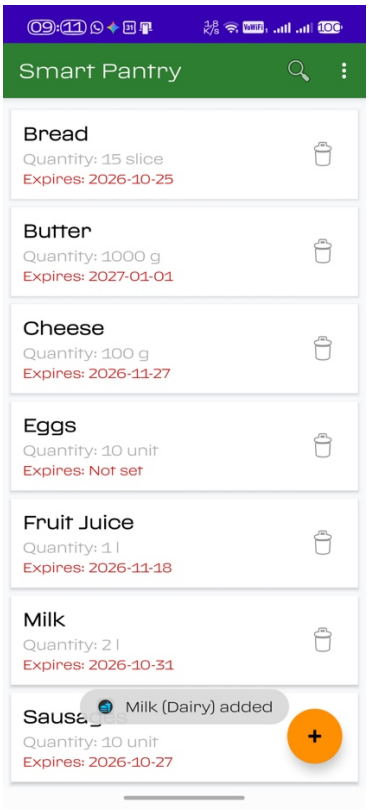


Figure 4:PantryListActivity. You can view all ingredients saved in the pantry. you can add with the "+" button or delete with the trash button. Also note the toast notification that shows up when adding or removing ingredients

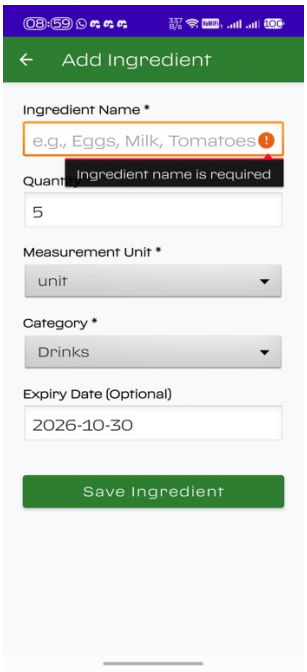


Figure 5: add ingredients activity. Note how it requires all ingredients to have a name

Code Snippets

Snippet 1: RecipeMatcher

```
for (PantryItem item : pantryItems) {
    if (item == null || item.getName() == null) {
        continue;
    }

    String normalizedPantryName = normalizeName(item.getName());
    String normalizedPantryUnit = normalizeUnit(item.getUnit());

    // Match ingredient name
    if (normalizedRequiredName.equals(normalizedPantryName)) {
        // Match unit or allow compatible unit definitions
        if (areUnitsCompatible(normalizedRequiredUnit, normalizedPantryUnit))
{
            totalQuantityFound += item.getQuantity();
        }
    }
```

This snippet will check every item stored in the pantry. It normalizes names & units to ensure they can be properly compared. Once normalized, it checks if the required ingredients are in the pantry list.

Snippet 2: DBHelper

```
@Override
public void onCreate(SQLiteDatabase db) {
    db.execSQL(CREATE_TABLE_PANTRY);
    db.execSQL(CREATE_TABLE_RECIPE);
    db.execSQL(CREATE_TABLE_RECIPE_INGREDIENT);
    seedDefaultRecipes(db);
}
```

onCreate is a method run the first time the database tries to get accessed. It's basically used to create a new database if one does not exist already. It creates the necessary tables. The method seedDefaultRecipes contains SQL code that inserts placeholder entries into the database.

Snippet 3: pantrylistactivity

```
protected void onResume() {
    super.onResume();
    loadPantryItems();
}
```

Whenever you return to the PantryListActivity, items in the list are reloaded. This ensures the list is up to date after the user adds or modifies the items.

Challenges and Solutions

Challenge 1: Linking git

The standard method to link android studio to git is to generate a token on GitHub, then paste it in the provided space in studio. You then have to pass your email and password. The issue is that I could not get android studio to accept my password. The solution was to use the terminal inside git to manually configure this, then create an empty git and push it. Only then could I get it to work properly.

Challenge 2: app icon creation

I wanted to create a custom app icon for the smart pantry app. The problem is that I got an error once I tried to build the final app. It turned out I had to move the icon foreground and background xml files to the drawable folder. After that I had to make sure the correct path was typed down inside the android manifest file.

Challenge 3: Text readability

When I ran my app, the white background blended in with the white text. I deleted the night theme and then manually configured the text color so that it contrasts with the background.

Conclusion & Reflection

Throughout this assignment we built a full android pantry app. The app contains helper classes, a database, and 5 activity screens that the user can interact with. We learnt how to design database entities and their relationships, updating properties, handling activity lifecycles and more. Challenges were faced like linking files within the app, updating the visual aspects, and linking the project to GitHub.

Future enhancements

We can add a filtering system to allow users to filter the ingredient list by category or expiration date. We can also do sorting.

We can add an image function so that every pantry has an associated image which makes it easier for users to find specific items.

We can build a proper notification system that alerts users more effectively of expiring goods. We can even have it suggest recipes to make on random days to help use up ingredients that are expiring soon.

Custom recipe function. Allow the user to create and store recipes in the app.

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